

Term Project Assignment

Your task for the term project assignment is to develop forecast models for State of Texas Sales Tax Receipts. You will be assigned to a 4-student team. The main data for the project is available on the class website in a ZIP file that includes and Excel workbook “Texas Tax Revenue.xlsx” with monthly data covering the period September 2003– February 2025 (more info below), a Team Assignment workbook, and a set of R code to ‘adjust’ sales tax revenue (more below). You will need to collect and incorporate other data/variables into your analysis.

These data are from the Texas Comptroller’s Office, and you may find some useful information on their website: <https://comptroller.texas.gov/transparency/revenue/watch/> , in particular, the Comptroller’s forecast of State revenue for the 2024-2025 – 2026-2027 fiscal years. Other resources on the comptroller site may also be helpful.

The “Texas Tax Revenue” workbook includes a tab “Sales Tax” where I have compiled the Sales Tax Receipts data into and time series format. In addition, there are separate tabs for fiscal years 2004 (Sept. 2023 – Aug. 2024) through 2024 that include other sources of State tax revenue. A time series plot of the Sales Tax variable will exhibit several significant single month drops in sales tax collections beginning in 2018. These relate to annual sales tax ‘holidays.’ The R code included in the ZIP file provides code to create an adjusted Sales Tax Receipts series that nets out the most dramatic effects of these holidays. I suggest that you use the adjusted series for your forecasting models and, if necessary, adjust your final forecasts to account for (average) revenue losses associated with the holidays.

Your final project submission should be compiled in a formal report format and should include the following elements:

1. Overview of the Sales Tax variable: Present a graphical analysis of the Sales Tax variable addressing the following elements, including whether any log-transformation is necessary for modelling purposes
 - a. Trend elements
 - b. Seasonal elements
 - c. Cyclical elements
2. Univariate Forecasting Models
 - a. Present your preferred ETS and ARIMA forecasting models for the Sales Tax variable based on a six-month “hold-out” sample where you forecast into the last six months of the data (that is, define your “training” set as ending in August 2024).
 - i. Present a basic model description for your preferred ETS and ARIMA models along with residual diagnostics.
 - ii. Present a six-month ahead forecast for September 2024 – February 2024. How do your forecasts compare to the actual values for the Sales Tax variable?
 - iii. Based on your analysis in i-ii, what is your preferred forecasting model, ETS or ARIMA?
 - b. Using the full dataset and the model you developed in 2.a., present a 30 month ahead path forecast (forecasts for each month) through the end of fiscal year 2027 in August 2027.
 - c. How do your forecasts (including any adjustments for tax ‘holidays’) compare to the Comptroller’s Sales tax forecasts for FY 2025, 2026, and 2027 (note that FY 2025 includes actual data for Sept. 2024 – Feb, 2025).
3. Multivariate Forecasting Model
 - a. Present a multivariate forecasting model based on a VAR model of the Sales Tax variable plus 2 or more series of your choice. These series could include either State or National level variables that may co-move with Texas Sales Tax receipts.
 - b. Your presentation should include a description of why you chose certain variables for your multivariate model and any pre-testing you undertook to select variables.
 - c. Based on your final multivariate model, present a 30 month ahead path forecast of the Sales Tax variable. How does it compare to your Univariate model forecasts and the Comptroller’s forecasts?

Organize your results into a formal report for submission in Canvas due by midnight **Wednesday, April 30th**.